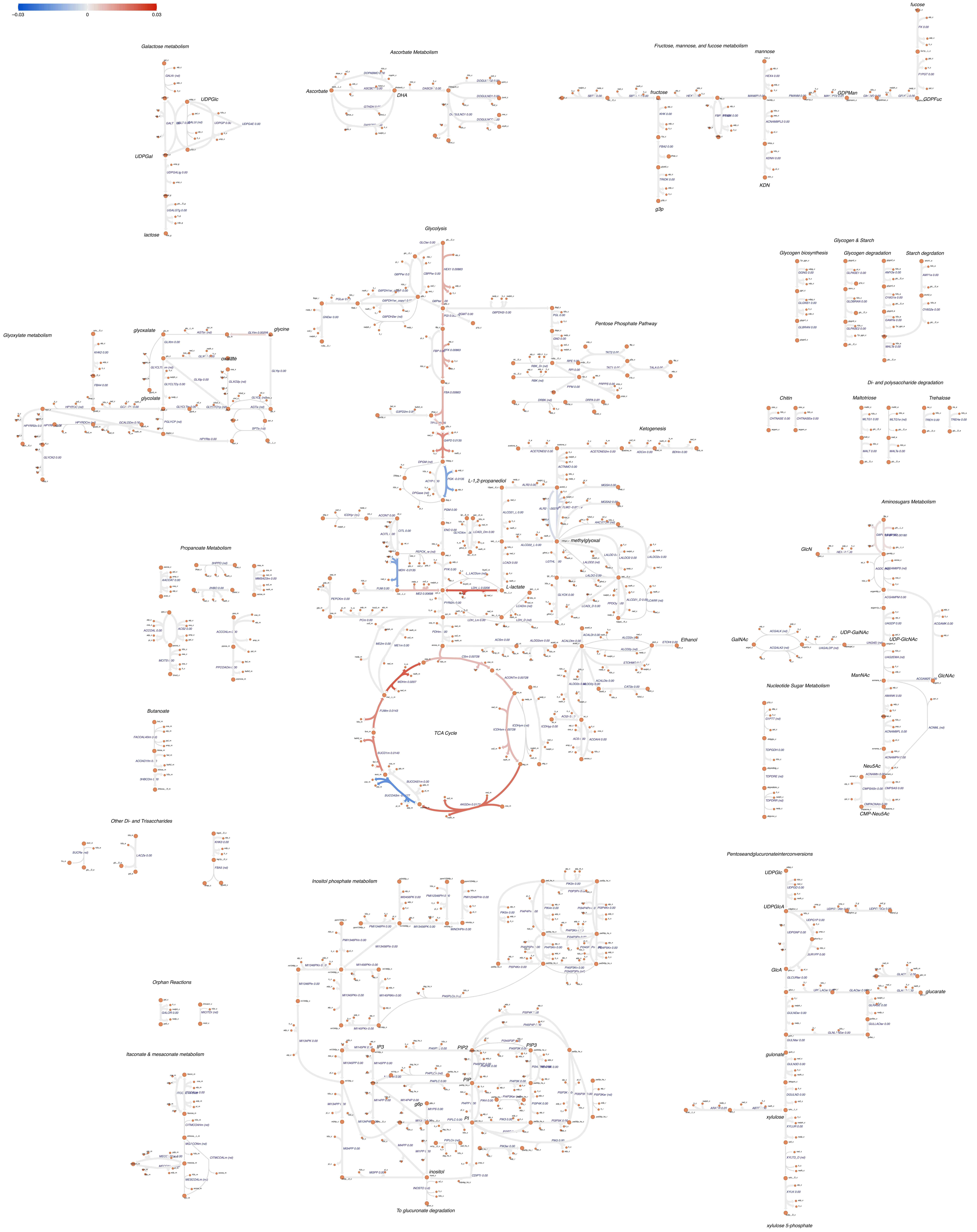
−0.03
0
0.03
Galactose metabolism
GALKr (nd)
UDPGlc
GALT
UGLT 0.00
GALUi (nd)
UDPGP 0.00
UDPG4E 0.00
UDPGal
UDPGALtg 0.00
UGALGTg 0.00
lactose
Ascorbate Metabolism
Ascorbate
DOPABMO 0.00
ASCBOX 0.00
DHA
DASCBH 0.00
GTHDH 0.00
DASCBR 0.00
DOGULND2 0.00
DOGULND1 0.00
DOGULNO1 0.00
DOGULNO2 0.00
Fructose, mannose, and fucose metabolism
mannose
fructose
SBTD 0.00
SBTD_D2 0.00
HEX7 0.00
KHK 0.00
FBA2 0.00
TRIOK 0.00
g3p
FBP26
PFK26 0.00
HEX4 0.00
MAN6PI 0.00
ACNAM9PL2 0.00
KDNH 0.00
KDN
PMANM 0.00
MAN1PT2 0.00
GDPMan
GMAND 0.00
GFUCS 0.00
GDPFuc
fucose
FK 0.00
F1PGT 0.00
Glycolysis
GLCter 0.00
HEX1 0.00863
G6PPer 0.00
CBPPer 0.00
G6PDH1er_copy1 0.00
G6PDH1er_copy1 0.00
G6PDH2er (nd)
PGLer 0.00
GNDer 0.00
G6Pter 0.00
PGI 0.00
PGMT 0.00
G6PDH2r 0.00
FBP 0.00
PFK 0.00863
FBA 0.00863
G3PD2m 0.00
TPI 0.00135
GAPD 0.0135
DPGM (nd)
PGK -0.0135
ACYP 0.00
DPGase (nd)
PGM 0.00
ENO 0.00
PYK 0.00
L-1,2-propanediol
ICDHyr (nd)
ACONT 0.00
ACITL 0.00
CITL 0.00
PEPCK_re (nd)
MDH -0.0135
FUM 0.00
ME2 0.00698
LDH_L 0.0206
L-lactate
PEPCKm 0.00
PYRt2m 0.00
PCm 0.00
LDH_Lm 0.00
PDHm 0.00
ME2m 0.00
ME1m 0.00
ACSm 0.00
CSm 0.00728
ACONTm 0.00728
ICDHym (nd)
ICDHxm 0.00728
MDHm 0.0207
FUMm 0.0143
TCA Cycle
SUCD1m 0.0143
SUCOAS1m 0.00
SUCOASm -0.0177
AKGDm 0.0177
GLYOXm 0.00
LCADi_Dm 0.00
L_LACDcm (nd)
Pentose Phosphate Pathway
PGL 0.00
GND 0.00
RPE 0.00
RPI 0.00
RBK_Dr (nd)
RBK (nd)
PPM 0.00
TKT2 0.00
TKT1 0.00
TALA 0.00
PRPPS 0.00
DRBK (nd)
DRPA 0.00
Ketogenesis
ACETONEt2 0.00
ACETONEt2m 0.00
ADCim 0.00
BDHm 0.00
ACTNMO 0.00
ALR3 0.00
MGSA 0.00
MGSA2 0.00
ALR2 -0.00279
ACTLMO -0.00279
AACTOOR (nd)
ALCD21_L 0.00
ALCD22_L 0.00
methylglyoxal
LCADi 0.00
LGTHL 0.00
LALDD 0.00
LALDD2 (nd)
LALDO2 0.00
LALDO2x 0.00
LALDO 0.00
GLYOX 0.00
LCADi_D 0.00
PPDOy 0.00
ALCD21_D 0.00
LCARR (nd)
LCADm (nd)
LDH_D (nd)
ALDD2xm 0.00
ACALDtm 0.00
ACALDt 0.00
ALCD2x (nd)
ALCD2y (nd)
ETOHt 0.00
Ethanol
ETOHMO 0.00
ALDD2x 0.00
ALDD2y 0.00
ACALDtx 0.00
CAT2p 0.00
ACt2r 0.00
ICDHyp 0.00
ACS 0.00
ACOAHi 0.00
Glyoxylate metabolism
glyoxalate
AGTim (nd)
GLYtm 0.00298
glycine
GLXtm 0.00
KHK2 0.00
FBA4 0.00
GLXO1 0.00
oxalate
GLYCLTDym (nd)
GLYCLTDy 0.00
GLXtp 0.00
GLXO2p (nd)
GLYtp 0.00
glycolate
HPYRDC (nd)
GCALDD 0.00
GLYCLTtp 0.00
GLYCTO1p (nd)
GLYOp (nd)
AGTix (nd)
HPYRR2x 0.00
HPYRRy 0.00
HPYRDCm (nd)
GCALDDm 0.00
PGLYCP (nd)
SPTix (nd)
HPYRtp 0.00
GLYCK2 0.00
Propanoate Metabolism
AACOAT 0.00
3HPPD (nd)
MMSAD3m 0.00
2HBO 0.00
ACCOAL 0.00
ACS2 0.00
ACCOALm 0.00
MCITS 0.00
PPCOAOm 0.00
Butanoate
FACOAL40im (nd)
ACOAD1fm 0.00
3HBCDm 0.00
Other Di- and Trisaccharides
SUCRe (nd)
LACZe 0.00
KHK3 0.00
FBA5 (nd)
Orphan Reactions
GALOR 0.00
MICITDr (nd)
Itaconate & mesaconate metabolism
ITCOAL1m 0.00
ITCOALm 0.00
CITMCOAHm (nd)
MGACONm (nd)
MECOAS1m 0.00
MECOAHm 0.00
CITMCOALm (nd)
MESCOALm (nd)
Glycogen & Starch
Glycogen biosynthesis
GGNG 0.00
GLGNS1 0.00
GLBRAN 0.00
Glycogen degradation
GLPASE1 0.00
GLDBRAN 0.00
GLPASE2 0.00
AMY2e 0.00
O16G1e 0.00
GAMYe 0.00
MALTe 0.00
Starch degrdation
AMY1e 0.00
O16G2e 0.00
Di- and polysaccharide degradation
Chitin
CHTNASE 0.00
CHTNASEe 0.00
Maltotriose
MLTG1 0.00
MLTG1e (nd)
MALT 0.00
MALTe 0.00
Trehalose
TREH 0.00
TREHe 0.00
Aminosugars Metabolism
G6PDA 0.00
GF6PTA 0.00180
GlcN
HEX10 0.00
AGDC (nd)
ACGAM6PS (nd)
ACGAMPM 0.00
UAGDP 0.00
ACGAMK 0.00
GalNAc
ACGALK (nd)
ACGALK2 (nd)
UAGALDP (nd)
UDP-GalNAc
UAG4Ei (nd)
UDP-GlcNAc
UAG2EMA (nd)
ManNAc
ACGAM2E 0.00
GlcNAc
AMANK 0.00
ACNAM9PL 0.00
ACNAMPH 0.00
ACNML (nd)
Neu5Ac
ACNAMtn 0.00
CMPSASn 0.00
CMPSAS 0.00
CMPACNAtn 0.00
CMP-Neu5Ac
Nucleotide Sugar Metabolism
G1PTT (nd)
TDPGDH 0.00
TDPDRE (nd)
TDPDRR (nd)
Pentoseandglucuronateinterconversions
UDPGlc
UDPGD 0.00
UDPGlcA
UDPGLCAtg 0.00
UDPGLDCg 0.00
UDPG1P 0.00
UDPGNP 0.00
GUR1PP 0.00
GlcA
GLCURter 0.00
GLAC 0.00
URIDLACer 0.00
GLACter 0.00
GLAC 0.00
glucarate
GLRASE 0.00
GULNDer 0.00
GULLACter 0.00
GLNLASEer 0.00
GULNter 0.00
gulonate
GULN3D 0.00
DGULND 0.00
ARABR 0.00
ABTD 0.00
xylulose
XYLUR 0.00
XYLTD_D (nd)
XYLK 0.00
xylulose 5-phosphate
Inositol phosphate metabolism
MI3456PK 0.00
PMI12346PH 0.00
PMI12346PHn 0.00
PMI1346PH 0.00
MI13456PK 0.00
MINOHPtn 0.00
PMI1346PHn 0.00
MI13456Ptn 0.00
MI1346PKn 0.00
MI1456PKn 0.00
MI1346Ptn 0.00
MI1345PKn 0.00
MI145P6Kn 0.00
MI145PKn 0.00
PI45PLCn (nd)
MI134PK 0.00
MI145PK 0.00
IP3
MI1345PP 0.00
MI145PP 0.00
PI45PLC 0.00
PIP2
MI14Ptn 0.00
PI4PLCn (nd)
PI4PLC 0.00
PIP
MI134PP 0.00
MI14PP 0.00
MI14P4P 0.00
g6p
MI1PS 0.00
MI134P4P 0.00
MI13PP 0.00
PIPLC 0.00
PI
PIPLCn (nd)
MI34PP 0.00
MI4PP 0.00
MI1PP 0.00
MI3PP 0.00
inositol
CDIPTr 0.00
INOSTO (nd)
To glucuronate degradation
PIK3n 0.00
PI3P3Pn 0.00
PIK5n 0.00
PI4P4Pn 0.00
PIK4n 0.00
PI34P4Pn 0.00
PI3P4Kn 0.00
PI4P3Kn 0.00
PI34P3Pn 0.00
PI45P5Pn 0.00
PI4P5Kn 0.00
PI345P5Pn 0.00
PI34P5Kn 0.00
PI5P4Kn 0.00
PI45P3Kn 0.00
PI345P3Pn (nd)
PI5P4K 0.00
PI45P4P 0.00
PI345P3P (nd)
PI45P3K 0.00
PIP3
PI45P5P 0.00
PI4P5K 0.00
PI34P5K 0.00
PI4P3K 0.00
PIK4 0.00
PI4P3Ker (nd)
PI4PP 0.00
PI3P4K 0.00
PI5P3K 0.00
PI35P3P 0.00
PI5P3Ker (nd)
PIK3 0.00
PI3P5K 0.00
PIK5 0.00
PIK3er 0.00